

KIRKWALL, United Kingdom

WMO: 030170

Lat: 58.9536N

Lon: 2.9014W

Elev: 51

StdP: 14.67

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.9	31.7	22.8	17.2	33.6	24.7	18.8	34.8	47.4	41.1	43.0	42.2	11.9	260	0.836

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	64.8	59.4	62.7	57.9	61.0	56.8	60.6	63.7	59.1	61.5	57.8	60.0	12.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	75.1	62.0	57.8	71.4	60.3	56.7	68.6	59.0	26.8	63.8	25.8	61.7	25.0	60.1	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.1	33.2	29.8	DB	26.4	70.3	3.4	2.6	24.0	72.1	22.0	73.7	20.1	75.1	17.7	77.0	(4)
(5)			WB	24.7	64.0	3.2	2.0	22.5	65.5	20.6	66.6	18.8	67.8	16.5	69.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	40.0	39.9	41.3	44.3	47.6	51.9	55.5	55.7	53.1	48.8	44.2	40.9	(6)
(7)		DBStd	6.86	3.56	3.96	4.05	3.98	3.84	3.33	3.17	2.97	3.35	3.82	4.19	4.48	(7)
(8)		HDD50	1686	309	282	270	175	93	18	1	1	12	67	177	281	(8)
(9)		HDD65	6579	774	702	735	622	539	394	294	287	358	503	625	746	(9)
(10)		CDD50	585	0	0	1	4	19	75	172	179	104	29	2	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	15.3	17.9	18.1	17.1	15.7	13.6	13.3	12.0	12.3	14.1	16.2	16.9	16.8	(14)
(15)	Precipitation	PrecAvg	40.0	4.1	3.3	3.4	2.2	1.9	2.1	2.2	2.9	3.7	4.4	5.0	4.4	(15)
(16)		PrecMax	53.5	7.0	6.9	6.6	4.5	3.5	4.3	4.2	7.6	6.9	9.3	8.5	8.7	(16)
(17)		PrecMin	29.3	1.5	0.7	1.0	0.4	0.2	0.4	0.7	0.7	1.3	1.3	1.9	1.6	(17)
(18)		PrecStd	5.2	1.3	1.5	1.3	0.9	0.8	0.8	0.9	1.3	1.5	1.7	1.6	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	51.4	53.8	57.7	62.7	65.0	68.8	68.0	63.9	58.8	54.4	52.3	(19)
(20)			MCWB	48.4	49.0	48.7	52.2	56.3	59.0	62.3	61.9	58.4	55.7	52.3	50.4	(20)
(21)		2%	DB	48.4	49.5	50.3	54.2	58.4	61.3	65.2	64.4	61.0	56.5	52.7	50.5	(21)
(22)			MCWB	46.6	47.7	47.0	49.5	52.8	55.8	60.0	59.3	56.7	53.8	51.0	48.9	(22)
(23)		5%	DB	46.8	47.6	48.6	51.9	55.8	59.3	62.8	62.3	59.3	55.2	51.6	49.0	(23)
(24)			MCWB	45.3	45.7	46.0	48.2	51.2	54.8	58.2	57.8	55.9	53.0	50.0	47.5	(24)
(25)		10%	DB	45.6	46.0	47.2	49.8	53.7	57.4	61.0	60.6	57.9	54.0	50.3	47.5	(25)
(26)			MCWB	44.1	44.1	44.9	46.8	49.7	53.5	57.1	56.9	55.2	52.0	48.7	46.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	49.9	50.0	53.1	57.0	59.7	63.5	62.6	59.8	56.4	53.0	51.0	(27)
(28)			MCDB	49.8	50.7	52.2	56.5	61.5	63.7	67.3	66.6	62.3	57.9	53.7	51.9	(28)
(29)		2%	WB	47.0	48.2	47.9	50.4	53.8	57.3	61.0	60.4	58.0	54.7	51.5	49.4	(29)
(30)			MCDB	48.0	49.4	49.5	53.2	57.0	60.2	64.0	63.2	59.7	55.8	52.3	50.4	(30)
(31)		5%	WB	45.6	46.1	46.6	48.6	52.0	55.7	59.3	58.9	56.8	53.6	50.3	47.7	(31)
(32)			MCDB	46.7	47.4	48.3	50.9	54.8	58.4	61.8	61.2	58.3	54.7	51.3	48.8	(32)
(33)		10%	WB	44.3	44.4	45.2	47.4	50.6	54.2	57.9	57.7	55.8	52.6	49.0	46.2	(33)
(34)			MCDB	45.6	45.8	46.9	49.4	53.0	56.6	60.2	59.6	57.4	53.8	50.2	47.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	7.2	8.4	9.1	10.3	9.5	9.5	9.4	8.6	7.6	6.9	6.9	(35)
(36)			MCDBR	7.3	8.2	10.0	11.3	12.8	12.1	12.4	11.5	10.1	8.0	7.5	7.9	(36)
(37)			MCWBR	7.2	7.3	7.6	7.5	8.1	7.8	7.9	7.3	7.1	6.9	7.5	8.0	(37)
(38)		5% WB	MCDBR	7.1	7.7	9.5	10.3	12.0	11.3	11.5	10.4	8.9	7.4	7.4	7.7	(38)
(39)			MCWBR	7.2	7.4	7.9	7.4	8.1	8.0	8.0	7.3	6.9	7.2	7.7	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.348	0.366	0.385	0.374	0.371	0.380	0.369	0.360	0.346	0.327	0.309	(40)	
(41)		taud	2.152	2.262	2.310	2.302	2.396	2.442	2.428	2.464	2.469	2.370	2.204	2.098	(41)	
(42)		Ebn at Noon	143	208	243	259	272	274	268	263	246	208	143	106	(42)	
(43)		Edh at Noon	16	23	29	35	34	33	32	29	25	21	15	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg	101	286	635	1120	1537	1531	1429	1159	747	377	138	64	(44)	
(45)		RadStd	10	30	48	89	105	122	105	83	60	34	12	7	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.49	N/A	+1.46	N/A	N/A	N/A	-142	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page